



Cite this: DOI: 10.1039/d0ay00493f

The selective and sensitive detection of formaldehyde by ZIF-90-LW via aza-Cope rearrangement†

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Formaldehyde (FA), as one of the simplest reactive carbonyl species (RCS), is widely known as an environmental toxin and carcinogen. In this work, a new ZIF-90 type material (ZIF-90-LW) was synthesized and investigated, which combines the two strategies of "2-aza-Cope rearrangement" and "MOF structure", by the combination of a pre-functionalized 2-allylaminoimidazole ligand and Zn²⁺ salt under solvothermal conditions. From this, the hurdle of selectivity over other carbonyl compounds (RCS) could be overcome despite their similar electrophilic reactivities to FA, and a prominent fluorescence turn-on type signal was realized through the 2-aza-Cope rearrangement mechanism. A good linear relationship ($R^2 = 0.9979$) was obtained by fitting the fluorescence intensity towards FA from 0 to 25 mM, and the detection limit of ZIF-90-LW for FA was 2.3 μ M. In addition, it also showed potentially useful sensing ability for the detection of FA in the gas phase, and might therefore be used to rapidly detect FA with a response time of 28 s in the liquid phase. All of the above features clearly demonstrate that ZIF-90-LW has great potential for sensitive and selective recognition of FA in the environment.

Received 8th March 2020

Accepted 19th June 2020

DOI: 10.1039/d0ay00493f

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1 Introduction

In recent years, reactive carbonyl compounds (RCS), traditionally considered as carcinogenic pollutants, have attracted widespread attention.^{1,2} Formaldehyde (FA), the simplest aldehyde, is widely used in the chemical industry for producing detergents and plastics, in wood processing, as a corrosion remover, in drug synthesis and so on. In addition, it is one of the endogenous aldehydes and is a highly reactive molecule that can be produced by metabolism in mammalian cells.^{3,4} Aldehydes are considered to be very dangerous carcinogens because of the formation of aldehyde-DNA addition products in the body.⁵ The normal concentration of FA in human blood is 0.06–0.08 mM,⁶ and a higher level of FA will cause many diseases, such as Alzheimer's disease,⁷ liver disease,⁸ allergic pulmonary inflammation,⁹ cardiovascular disease¹⁰ and cancer.¹¹

Considering the factors mentioned above, the detection of FA is imminent, and various detection methods have been developed in recent years. Conventional FA detection methods,

including spectrophotometry,¹² colorimetry,¹³ gas chromatography,¹⁴ high-performance liquid chromatography,¹⁵ ion chromatography,¹⁶ and polarography,¹⁷ are time consuming and require expensive test equipment and technicians. On the other hand, fluorescent probes provide several advantages for FA detection technology because of their sensitivity, rapidity, lower detection limit, and simple operation. So, effective detection methods for FA recognition based on the fluorescence technique have been reported to date.^{18–33}

So far, several different but complementary mechanisms of FA detection have been reported, and mainly involve (Table 1): (a) the condensation of free amino groups ($-\text{NH}_2$) with FA to form the formimine-based detection methods;^{18,19} (b) the condensation of FA and amines to form the aminal-based detection methods;^{20–24} (c) the use of a homoallylamine with a FA selective trigger for FA reactivity with a 2-aza-Cope rearrangement reaction as shown in Table 1.^{2a,25–33} The alternative reactivity-based approaches of (a) and (b) exploit the formimine- or aminal-forming properties between FA and amines, and both display fast reaction kinetics but it is more challenging to tune for FA selectivity over other aldehydes. Thus, they offer distinct advantages and disadvantages over the 2-aza-Cope strategy. So, (a), (b) and (c) offer unique advantages and disadvantages, as summarized in Table 2.

Metal-organic frameworks (MOFs), as a new type of porous coordination compound in the past few decades, have received enormous attention in applications of electricity, light and magnetism compared with traditional porous materials.³⁹ They

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† Electronic supplementary information (ESI) available. See DOI: 10.1039/d0ay00493f

Table 1 Molecular probe strategies for the detection of FA

Response mechanism		Schematic diagram	
a)	Formimine-based detection		
b)	Aminal-based detection		
c)	2-Aza-Cope-based detection		

have great potential in many frontier directions related to environmental chemistry, such as catalysis,^{40,41} gas storage, gas adsorption and separation,⁴² gas sensing,^{43,44} heavy metal ion purification and so on. At the same time, various luminescent MOFs have been widely developed for detecting ions, gas molecules, solvent molecules, dye molecules, explosive molecules *etc.*, due to their properties of porosity, tunable pore size, and functionalized inner and outer surfaces.⁴⁵ Based on them being an ideal platform to fabricate fluorescent probes for the detection of environmental pollutants, some MOFs have also been used for the detection of FA, such as UIO-66-NH₂,³⁵ Al-MIL-53-N₂H₃,³⁶ MOF-5,⁴⁶ [Zn(NH₂bdc)(bix)]_n,⁴⁷ γ-CD-MOF-K,⁴⁸ and so on.

In principle, organic fluorophores can be grafted on MOF frameworks *via* either pre-synthetic modification or post-synthetic modification. Thus, herein, we designed a novel ZIF-90 type of MOF, namely, **ZIF-90-LW**, which combines two strategies of “2-aza-Cope rearrangement” and “MOF structure” (**ZIF-90-LW**) through the combination of the pre-functionalized 2-allylaminimidazole ligand and Zn²⁺ salt under solvothermal conditions. Indeed, it was the 2-aza-Cope reaction that exhibited high selectivity for FA over other RCS and carbonyl species, such as acetaldehyde, and methylglyoxal. Consequently, the change in nucleophilicity and electronic properties on conversion of a homoallylamine to a carbonyl could be harnessed to provide a fluorescence turn-on change. The obtained **ZIF-90-LW**

with a homoallylamine imidazole-based moiety is a highly sensitive and selective MOF-based probe for detecting FA *via* fluorogenic routes (turn-on).

2 Experimental section

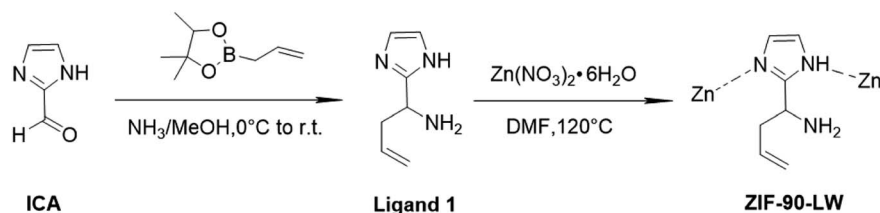
2.1. Materials and instrumentation

Imidazole-2-formaldehyde was purchased from J&K; allylboronic acid pinacol ester was purchased from Energy Chemical; zinc nitrate hexahydrate was purchased from Sinopharm Chemical Reagent Co., Ltd; ammonia (7 M in MeOH) was purchased from MERYER (Shanghai, China). All the other chemicals obtained from commercial sources were analytically pure and used without further purification, except for ZIF-90 which was synthesized by the literature method.⁴⁷ The water was purified by a Millipore filtration system.

Ultraviolet visible (UV) absorption spectra were obtained by a UV-2600 spectrometer (Techcomp., China). X-Ray diffraction (XRD) data were obtained with an Ultima IV multipurpose X-ray diffraction system (Japan). Fluorescence emission studies were carried out with a Hitachi F-4500 Fluorescence Spectrophotometer. Fourier transform infrared (FT-IR) spectra were recorded in the region of 440–4000 cm^{−1} with a FTIR-8400s Fourier Transform Infrared Spectrometer (Shimadzu). The scanning electron microscope (SEM) used was an Ultra-high Resolution Scanning Electron Microscope

Table 2 A comparison of the performances of various formaldehyde sensors

Probe	Type of material	Response mechanism	Linear range	L_{OD}	FA concentration for response time	Response time
RFFP ²⁵	Organic molecule	Turn-on	0–3 mM	18.7 mM	25 mM	>60 min
MQAP ²⁶		Turn-on	0–5 mM	4.054 μM	5.0 mM	—
HBT-FA ²⁸		Turn-on	0–15 mM	0.41 mM	20 mM	180 min
FAP-1 (ref. 29)		Turn-on	0–100 μM	5 μM	100 μM	120 min
PBD-FA ³⁴		Turn-on	0–1 mM	43.5 nM	0.8 mM	100 min
AENO ³³	MOF	Turn-on	0–5 mM	0.57 μM	1.0 mM	150 min
Uio-66-NH ₂ (ref. 35)		Turn-on	0.333–33.3 mM	0.13 mM	100 ppm	2 min
1' (ref. 36)		Turn-on	100–400 μM	8.3 μM	20 mM	1 min
Ag ⁺ /Eu ³⁺ -Uio-66-NH ₂ (ref. 37)		Turn-on	0–33.3 μM	1.7 μM	33.3 μM	6 min
Eu/Zr-Uio-66-NH ₂ (ref. 38)		Turn-on	—	6.7 μM	—	—
ZIF-90-LW		Turn-on	0–25 mM	2.3 μM	25 mM	28 s



Scheme 1 The synthesis of ZIF-90-LW.

SU8010. Thermogravimetric analysis (TG) data were obtained by a METTLER TOLEDO TGA/DSC1 Synchronous Thermal Analyzer. ^1H NMR spectra were measured on a Varian Gemini-400 MHz spectrometer with chemical shifts reported in ppm (TMS as an internal standard). Electrospray ionization (ESI) mass spectra were collected on Bruker Daltonics Inc. Pgeneral TU-1901 and Hitachi F-4500 fluorescence spectrometers were used in obtaining UV-Visible and fluorescence spectra at 25 °C. All chemical reactions were detected by thin layer chromatography under a 254 nm (or 365 nm) UV lamp. Silica gel plates (HSGF254, 0.2 mm) were purchased from Yantai Chemical Industry Research Institute. Silica gel plates (HSGF254, 0.2 mm) were purchased from Yantai Chemical Industry Research Institute.

In the fluorescence sensing experiments of **ZIF-90-LW**, a stock solution of FA and other analytes was prepared in ultrapure water, and a stock suspension of **ZIF-90-LW** (25 mM, 0.4 mg mL⁻¹) was prepared in DMF. The samples were excited at 380 nm with the excitation and emission slit widths set at 5.0 nm and 10.0 nm, respectively. The emission spectrum was scanned from 350 nm to 650 nm at 1200 nm min⁻¹.

2.2 Synthesis

The detailed synthetic procedure of **ZIF-90-LW** is outlined in Scheme 1.

2.2.1 Preparation of 2-allylaminoimidazole ligand (Ligand 1). 0.48 g (5 mmol) of imidazole-2-carboxaldehyde (ICA) was dissolved in 5 mL of MeOH at 0 °C, and then 7.2 mL of NH₃ solution (7 M in MeOH, 50 mmol) was added to the solution. The reaction mixture was stirred at 0 °C for 30 min, followed by the addition of allylboronic acid pinacol ester (1.68 g, 10 mmol). The reaction mixture was warmed to ambient temperature overnight. The solvent was removed under reduced pressure, and the resulting residue was purified by silica column chromatography using CH₂Cl₂/CH₃OH (v/v, 20 : 1) to afford a yellow solid (yield 68%).²⁹ ^1H NMR (400 MHz, CDCl₃) δ (TMS, ppm): 2.38–2.49 (q, 1H), 2.63–2.73 (q, 1H), 3.94–3.99 (t, 1H), 4.88–4.96 (d, J = 32 Hz, 1H), 5.13–5.24 (d, J = 44 Hz, 1H), 5.82–5.96 (q, 1H), 6.92–6.98 (d, J = 24 Hz, 1H), 7.02–7.08 (d, J = 24 Hz, 1H), 8.77 (s, 1H), 13.3 (s, 1H). ESI (+)-HRMS (m/z): for **Ligand 1**: calcd, 137.100, found: $[\text{M} + \text{H}]^+$, 138.102.

2.2.2 Synthesis of ZIF-90-LW. **ZIF-90-LW** was obtained by heating a solution of zinc nitrate hexahydrate (Zn(NO₃)₂·6H₂O) (0.62 g, 2.1 mmol) and **Ligand 1** (0.41 g, 3 mmol) in *N,N*-dimethylformamide (DMF, 10 mL) for 18 h at 100 °C. The mixture was cooled to room temperature and an orange solid

precipitated, and then the solids were filtered out. To activate the MOF material, the filtered solids were placed in a stainless steel autoclave and heated to 150 °C overnight using a high temperature furnace (yield 81% calculated based on **Ligand 1**, ESI 4†).

2.2.3 Synthesis of ZIF-90. For structural comparison, we synthesized ZIF-90 according to the methods reported.⁴⁹ ZIF-90 were obtained by heating a solution of Zn(NO₃)₂·6H₂O and imidazole-2-carboxaldehyde (ICA) in *N,N*-dimethylformamide (DMF, 3 mL) for 18 h at 100 °C. The product was collected by filtration. Unreacted raw material was washed with DMF several times and activated in the same way as **ZIF-90-LW** (yield 90% calculated based on ICA).

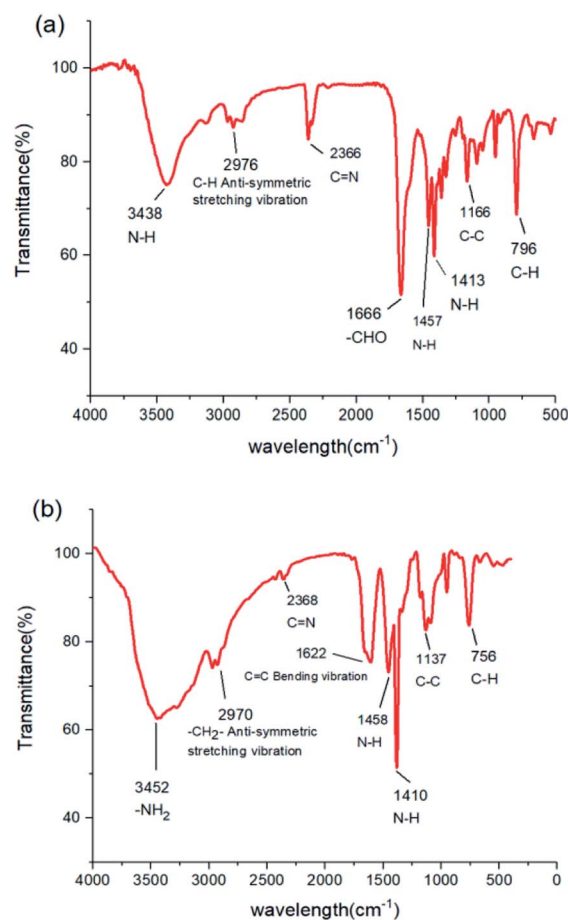


Fig. 1 (a) The FT-IR spectrum of ZIF-90. (b) The FT-IR spectrum of ZIF-90-LW.

3 Results and discussion

3.1 Characterization of the prepared fluorescent probe materials

Ligand 1 was prepared by imidazole-2-carboxaldehyde (ICA) and allylboronic acid pinacol ester in NH_3/MeOH solution. The obtained **Ligand 1** was used to synthesize **ZIF-90-LW** by heating a mixture of **Ligand 1** and $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ in DMF.

The morphological and structural characterizations of **ZIF-90-LW** were determined by the following tests. First, FT-IR measurements were carried out in order to prove the structure of **ZIF-90-LW**. As shown in Fig. S14a–c,† the stretching vibration peaks of the $-\text{NH}_2$ group and N–H in the imidazole ring of

Ligand 1 appeared at 3497, and 1417 and 1492 cm^{-1} , while the peaks around 3452, and 1410 and 1456 cm^{-1} of the **Ligand 1** became wide and narrow, and sharp respectively after coordination with Zn^{2+} .

From the comparison of the FT-IR spectra for ZIF-90 and **ZIF-90-LW** (Fig. 1a and b), it can be found that more absorption peaks appeared in the FT-IR spectrum of **ZIF-90-LW** than in ZIF-90, attributed to functional groups in the **Ligand 1** modified section (Fig. S14c†). The peak of the aldehyde group at 1666 cm^{-1} from the ICA ligand disappeared, and new peaks were generated at 3452 cm^{-1} ($-\text{NH}_2$) and 1622 cm^{-1} ($-\text{C}=\text{C}$), suggesting that the conversion of the starting material of **Ligand 1** into the resulting **ZIF-90-LW** was carried out. In particular, the similarity of **ZIF-90-LW** to ZIF-90 is that both of them have characteristic peaks of $\text{N}-\text{Zn}^{2+}$ at 1410–1458 cm^{-1} . Thus, **ZIF-90-LW** belongs to the same series as ZIF-90-MOF in morphology.

Besides, comparisons of the skeletal topology between ZIF-90, the simulated ZIF-90 and **ZIF-90-LW** powders were conducted by powder X-ray diffraction (PXRD). As can be seen from Fig. 2, ZIF-90 is consistent with the simulated ZIF-90 pattern, and all the main diffraction peaks of **ZIF-90-LW** and ZIF-90 at 4.0°, 10.0°, 12.5° and 16.3° showed great similarity to those of the simulated ZIF-90 pattern. The sharp diffraction peaks of the PXRD pattern indicated that not only was there an excellent crystallinity but also it showed the same crystallographic structure as the pristine ZIF-90.

The morphology characterizations of ZIF-90 and **ZIF-90-LW** were performed by scanning electron microscopy (SEM) as indicated in Fig. 3a–d. The morphologies showed that ZIF-90 has a very uniform morphology and a special cubic close-packed structure (Fig. 3a and b). Meanwhile, the obtained **ZIF-90-LW** featured an irregular and hexahedron morphology with a diameter of 300 nm, and still maintained a uniform cubic structure (Fig. 3c and d), only slightly different to ZIF-90.

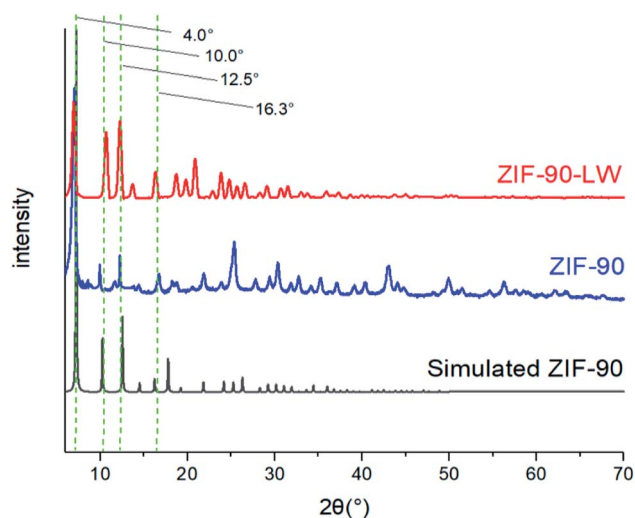


Fig. 2 PXRD data of ZIF-90-LW and ZIF-90, and the simulated pattern of ZIF-90.

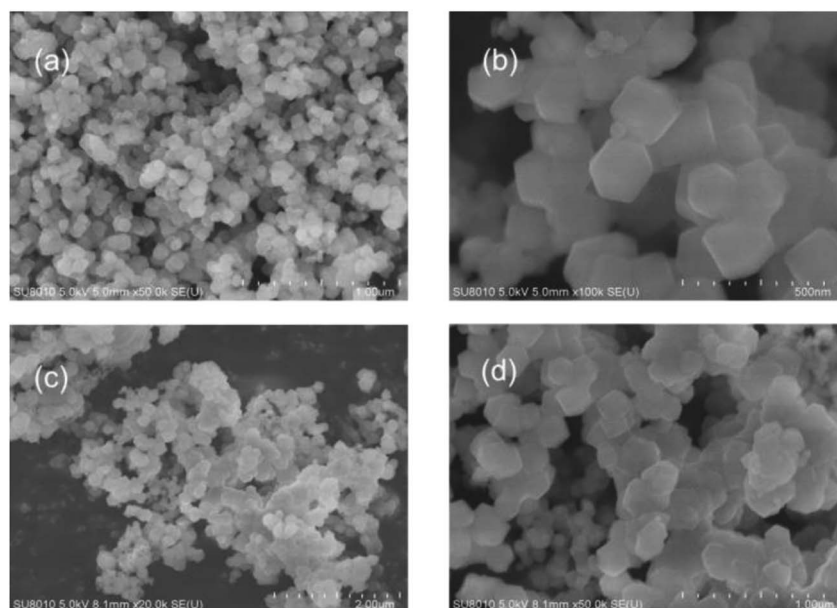


Fig. 3 (a and b) SEM images of ZIF-90. (c and d) SEM images of ZIF-90-LW.

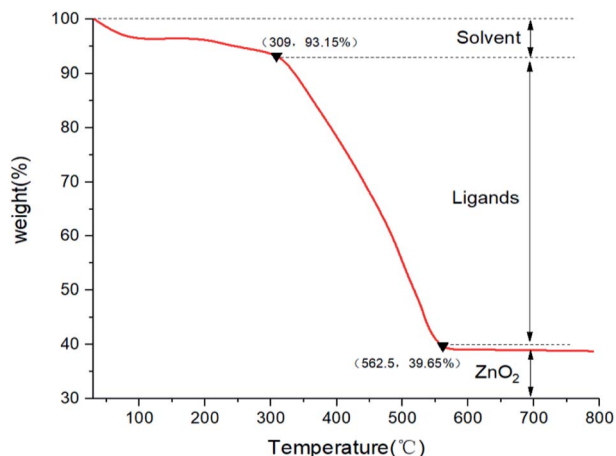


Fig. 4 Thermogravimetric analysis (TGA) of ZIF-90-LW.

Finally, thermogravimetric analysis (TGA) of ZIF-90-LW was carried out in an air atmosphere at a temperature range from 30–800 °C to study its thermal stability (Fig. 4). TGA showed that DMF in the MOF pores gradually evaporated at 0–309 °C, and ZIF-90-LW exhibited thermal stability up to 309 °C in air and gradually disintegrated in the interval of 309–562.5 °C. After 562.5 °C, the ligand molecules were separated from the skeleton and a residue of zinc dioxide remained.

3.2 Fluorescence sensing experiments

3.2.1 Effects of pH on the detection. The effects of pH for FA determination were explored. ZIF-90-LW (25 mM, 0.4 mg mL⁻¹) was added to test tubes that contained 25 mL of DMF and Tris buffer solution (7 : 3, v/v), and sonicated. We measured the fluorescence intensity of ZIF-90-LW in response to FA at different pHs from 1.0 to 14.0. The pH was adjusted with HCl and NaOH solution and monitored with a pH meter. The fluorescence results revealed that the best detection condition of ZIF-90-LW for FA was pH 7.0 (Fig. 5).

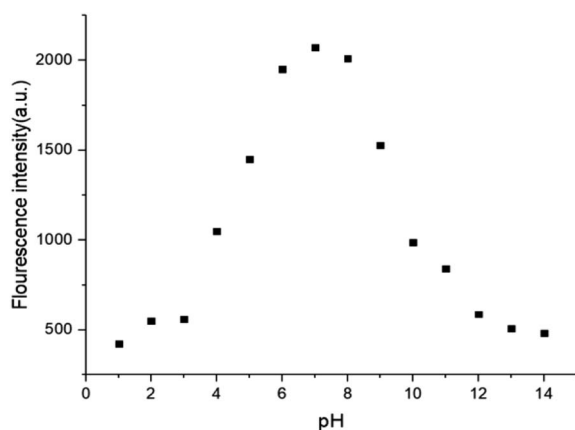


Fig. 5 The fluorescence intensity changes of ZIF-90-LW in response to FA at different pH values. ZIF-90-LW: 25 mM (0.4 mg mL⁻¹), FA: 25 mM, in a mixture of DMF and Tris buffer solution (7 : 3, v/v). λ_{ex} = 380 nm, λ_{em} = 438 nm.

3.2.2 Study of the fluorescence properties of ZIF-90-LW at different FA concentrations. In order to investigate the response of ZIF-90-LW to different concentrations of FA, concentration-dependent fluorescence titration experiments were performed. The fluorescence intensity of the suspension of ZIF-90-LW (25 mM, 0.4 mg mL⁻¹) increased gradually with the addition of FA at 438 nm, as shown in Fig. 6. Simultaneously, a perceived fluorescence color change of ZIF-90-LW was observed from colorless to blue (Fig. 6a, inset). A good linear relationship ($R^2 = 0.9979$) was obtained by fitting the fluorescence intensity towards FA (0 to 25 mM). Meanwhile, the detection limit was found to be 2.3 μM on the basis of the equation $L_{\text{OD}} = 3\delta/m$ (δ is the standard deviation of the blank solution and m is the absolute value of the slope between fluorescence intensity and fluoride ion concentration) (Fig. S4†).

3.2.3 Time-dependence of the fluorescence intensity.

Rapid response is a critical index for reactive-type fluorescent probes. For the time-dependent study, 500 μL of 25 mM FA was introduced to the suspension of ZIF-90-LW (25 mM, 0.4 mg mL⁻¹) in DMF and the fluorescence spectra were collected at

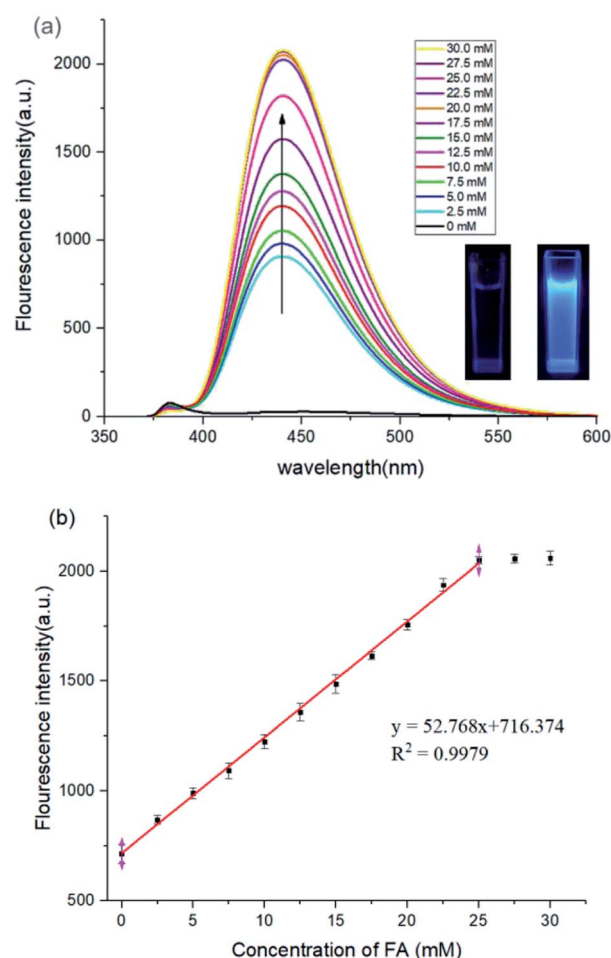


Fig. 6 (a) Fluorescence spectra of ZIF-90-LW with the addition of different concentrations of FA. (b) The linear relationship between the fluorescence intensity and FA concentration. ZIF-90-LW: 25 mM (0.4 mg mL⁻¹), FA: 0–30 mM, in a mixture of DMF and Tris buffer solution (7 : 3, v/v), pH = 7.0. λ_{ex} = 380 nm, λ_{em} = 438 nm.

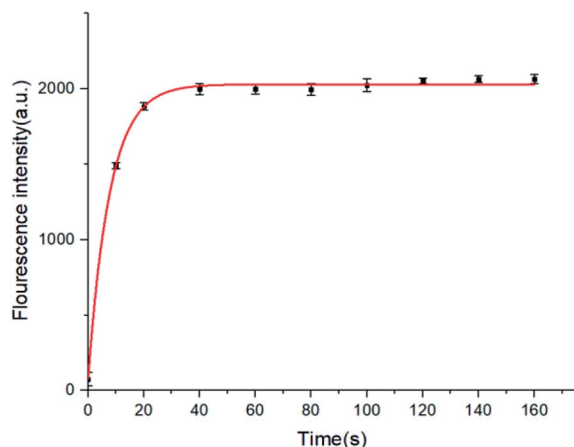


Fig. 7 A pseudo-first-order kinetics plot of the reaction of ZIF-90-LW with FA. ZIF-90-LW: 25 mM (0.4 mg mL^{-1}), FA: 25 mM, in a mixture of DMF and Tris buffer solution (7 : 3, v/v), pH = 7.0. $\lambda_{\text{ex}} = 380 \text{ nm}$, $\lambda_{\text{em}} = 438 \text{ nm}$.

regular intervals from 0 s to 160 s. As shown in Fig. 7, upon the addition of FA, the fluorescence intensity of ZIF-90-LW showed a notably increase as the reaction time increases. Then the fluorescence intensity of ZIF-90-LW reached the maximum value and reached a plateau within 28 s, indicating that the best response time was 28 s. Further kinetic studies also showed that the calculated pseudo-first-order rate constant (k') was 7.93 min^{-1} (Fig. S5†). All of the above results demonstrated that the probe ZIF-90-LW could detect FA in a quantitative manner and with a rapid response time by the fluorescence spectrometry method.

3.2.4 Selectivity of ZIF-90-LW toward FA. The specificity of ZIF-90-LW toward FA was investigated with various analytes, including oxaldehyde, butyraldehyde, valeraldehyde, benzaldehyde, acetone, propionaldehyde, ethyl acetate, hydrogen peroxide, and glucose, which demonstrated excellent selectivity of ZIF-90-LW to FA. As shown in Fig. 8, FA induced a large fluorescence increase of ZIF-90-LW, whereas other analytes triggered a small or negligible fluorescence variation. These

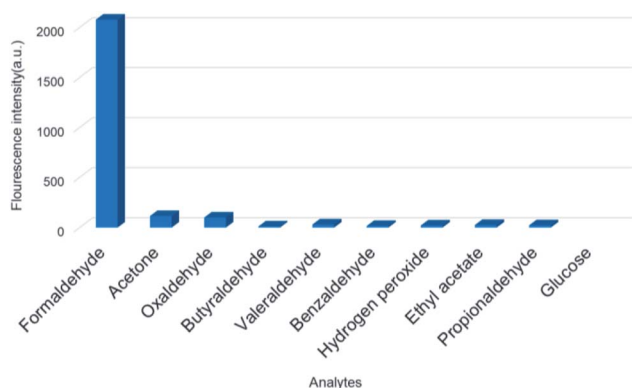


Fig. 8 Fluorescence intensity of ZIF-90-LW in different analytes. ZIF-90-LW: 25 mM (0.4 mg mL^{-1}), FA: 25 mM, in a mixture of DMF and Tris buffer solution (7 : 3, v/v), pH = 7.0. $\lambda_{\text{ex}} = 380 \text{ nm}$, $\lambda_{\text{em}} = 438 \text{ nm}$.

results indicated that ZIF-90-LW exhibits good sensitivity toward FA over other species at 438 nm, which suggested that ZIF-90-LW could be used as an effective probe for practical detection of FA.

3.3 Mechanism of FA sensing

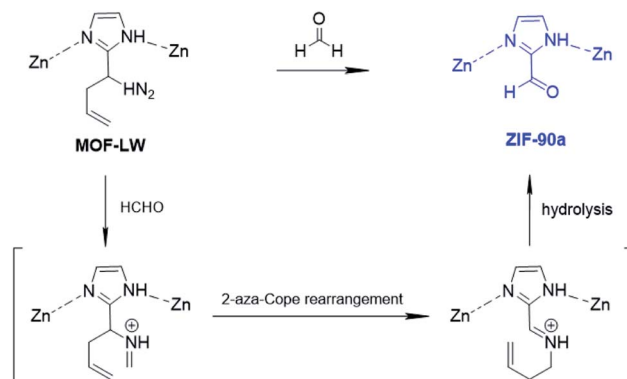
We supposed that once ZIF-90-LW reacted with FA, newly formed intermediates would rapidly undergo 2-aza-Cope rearrangement reaction^{25–33} to produce ZIF-90a with strong fluorescence emission spontaneously (Scheme 2). The change in the electron density of the recognizing product of ZIF-90a directly reflected the fluorescence property, which is an enhanced blue fluorescence because of the presence of the electron-efficient core.

To confirm this mechanism, PXRD and ESI mass spectra tests were conducted on ZIF-90a. As shown in Fig. S6,† the PXRD of ZIF-90a was consistent with the simulated pattern of ZIF-90. Subsequently, ZIF-90a was collapsed by soaking in HCl (pH = 1.0) to obtain the ICA ligand. A peak can be found at 96.031 in the ESI mass spectra (Fig. S7†), suggesting that ZIF-90-LW and FA reacted according to the above expected mechanism.

In addition, considering that FA is the smallest and most electrophilic aldehyde compared with other analytes and can easily pass through into the pores of the MOF structure and react with its functional groups, ZIF-90-LW can be utilized in the removal of FA and is a promising candidate for environmental monitoring of FA in real scenarios.

3.4 Application of fluorescence test paper of ZIF-90-LW

To detect FA gas, we assembled ZIF-90-LW onto test paper. Clean blank neutral filter paper was soaked in a ZIF-90-LW suspension (25 mM, 0.4 mg mL^{-1}) for 3 minutes, and then taken out and dried. This step was repeated three times. 10 mL of 25mM FA solution was added to a small bottle and then covered with a large beaker. After the FA gas was fully volatilized in the beaker, the six prepared test papers were placed in the beaker one by one, and kept for 0 s, 5 s, 10 s, 15 s, 20s and 28 s, respectively (Fig. S9†). The obtained six test papers were placed under an ultraviolet lamp (excitation wavelength $\lambda_{\text{ex}} = 356 \text{ nm}$) to show significant blue fluorescence, and the fluorescence



Scheme 2 Design showing the response of ZIF-90-LW towards FA.

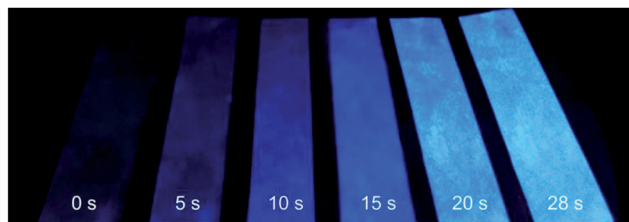


Fig. 9 The fluorescence response of ZIF-90-LW test paper after exposure to FA gas under a 365 nm UV lamp.

intensity increased with increasing time (Fig. 9). It is worth mentioning that the response time of FA gas was 28 s, which is much smaller than the response time of many other FA sensors.

4 Conclusions

In conclusion, a highly selective fluorescent probe towards FA based on a ZIF-90 type material (ZIF-90-LW) was synthesized and investigated. The new probe combined the two strategies of “2-aza-Cope rearrangement” and “MOF structure” (ZIF-90-LW), with the combination of a pre-functionalized 2-allylaminoimidazole ligand and Zn^{2+} salt under solvothermal conditions. In this way, even though other carbonyl compounds (FAS) have similar electrophilic reactivities, the selectivity barrier of other carbonyl compounds (RCS) can be avoided, and a prominent fluorescence turn-on type signal was realized via the 2-aza-Cope rearrangement mechanism. The detection limit of ZIF-90-LW for FA was 2.3 μM in the dynamic range of 0 to 25 mM. In addition, it also showed potentially useful sensing ability for the detection of FA in the gas phase, and it might therefore be used to rapidly detect FA with a response time of 28 s in the liquid phase. Therefore, the new functionalized MOF-type material of ZIF-90-LW is certainly promising for the sensitive and selective detection of FA. We believe that the work presented here may help to increase the readiness for FA detection in the environment in the case of an incident.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

This research was supported by the National Science Foundation of China (No. 21575012), and the Natural Science Foundation of Beijing (No. 2151003).

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